

Foundations of Curse of dimensionality

[Curse of dimensionality]

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} r^d$$

1.0 Content Block

Therefore, the squared distance from the origin, $r^2 = \sum_i x_i^2$ has the average value $d/3$ and variance $4d/45$. For large d , distribution of r^2/d is close to the normal distribution with the mean $1/3$ and the standard deviation $2/\sqrt{45d}$ according to the central limit theorem. Thus, when uniformly generating points in high dimensions, both the "middle" of the hypercube, and the corners are empty, and all the volume is concentrated near the surface of a sphere of "intermediate" radius $d/3$. This also helps to understand the chi-squared distribution. Indeed, the (non-central) chi-squared distribution associated to a random point in the interval $[-1, 1]$ is the same as the distribution of the length-squared of a random point in the d -cube. By the law of large numbers, this distribution concentrates itself in a narrow band around d times the standard deviation squared (σ^2) of the original derivation. This illuminates the chi-squared distribution and also illustrates that most of the volume of the d -cube concentrates near the boundary of a sphere of radius σd . A further development of this phenomenon is as follows. Any fixed distribution on the real numbers induces a product distribution on points in $\mathbb{R}^d$. For any fixed n , it turns out that the difference between the minimum and the maximum distance between a random reference point Q and a list of n random data points $P_1, \dots, P_n$ become indiscernible compared to the minimum distance:

2.0 Content Block

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings such as the three-dimensional physical space of everyday experience. The expression was coined by Richard E. Bellman when considering problems in dynamic programming. The curse generally refers to issues that arise when the number of datapoints is small (in a suitably defined sense) relative to the intrinsic dimension of the data. Dimensionally cursed phenomena occur in domains such as numerical analysis, sampling, combinatorics, machine learning, data mining and databases. The common theme of these problems is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse. In order to obtain a reliable result, the amount of data needed often grows exponentially with the dimensionality. Also, organizing and searching data often relies on detecting areas where objects form groups with similar properties; in high dimensional data, however, all objects appear to be sparse and dissimilar in many ways, which prevents common data organization strategies from being efficient.

3.0 Content Block

Concentration of scores and distances: derived values such as distances become numerically similar
 Irrelevant attributes: in high dimensional data, a significant number of attributes may be irrelevant
 Definition of reference sets: for local methods, reference sets are often nearest-neighbor based
 Incomparable scores for different dimensionalities: different subspaces produce incomparable scores
 Interpretability of scores: the scores often no longer convey a semantic meaning
 Exponential search space: the search space can no longer be systematically scanned
 Data snooping bias: given the large search space, for every desired significance a hypothesis can be found
 Hubness: certain objects occur

more frequently in neighbor lists than others. Many of the analyzed specialized methods tackle one or another of these problems, but there remain many open research questions.

4.0 Content Block

k-nearest neighbor classification Another effect of high dimensionality on distance functions concerns k-nearest neighbor (k-NN) graphs constructed from a data set using a distance function. As the dimension increases, the indegree distribution of the k-NN digraph becomes skewed with a peak on the right because of the emergence of a disproportionate number of hubs, that is, data-points that appear in many more k-NN lists of other data-points than the average. This phenomenon can have a considerable impact on various techniques for classification (including the k-NN classifier), semi-supervised learning, and clustering, and it also affects information retrieval.

5.0 Content Block

Machine learning In machine learning problems that involve learning a "state-of-nature" from a finite number of data samples in a high-dimensional feature space with each feature having a range of possible values, typically an enormous amount of training data is required to ensure that there are several samples with each combination of values. In an abstract sense, as the number of features or dimensions grows, the amount of data we need to generalize accurately grows exponentially. A typical rule of thumb is that there should be at least 5 training examples for each dimension in the representation. In machine learning and insofar as predictive performance is concerned, the curse of dimensionality is used interchangeably with the peaking phenomenon, which is also known as Hughes phenomenon. This phenomenon states that with a fixed number of training samples, the average (expected) predictive power of a classifier or regressor first increases as the number of dimensions or features used is increased but beyond a certain dimensionality it starts deteriorating instead of improving steadily. Nevertheless, in the context of a simple classifier (e.g., linear discriminant analysis in the multivariate Gaussian model under the assumption of a common known covariance matrix), Zollanvari et al. showed both analytically and empirically that as long as the relative cumulative efficacy of an additional feature set (with respect to features that are already part of the classifier) is greater (or less) than the size of this additional feature set, the expected error of the classifier constructed using these additional features will be less (or greater) than the expected error of the classifier constructed without them. In other words, both the size of additional features and their (relative) cumulative discriminatory effect are important in observing a decrease or increase in the average predictive power. In metric learning, higher dimensions can sometimes allow a model to achieve better performance. After normalizing embeddings to the surface of a hypersphere, FaceNet achieves the best performance using 128 dimensions as opposed to 64, 256, or 512 dimensions in one ablation study. A loss function for unitary-invariant dissimilarity between word embeddings was found to be minimized in high dimensions.